

Sharath Y Kudachi

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CAREER OBJECTIVE

Third-year Computer Science and Engineering student with a strong foundation in software development, data structures, algorithms and database management. Experienced in developing AI-powered applications, real-time monitoring systems, full-stack web applications and intelligent data processing solutions through academic projects, hackathons and research. Worked on autonomous UAV navigation and trajectory optimization using path planning and optimization algorithms, with exposure to API integration and scalable system design. Passionate about solving real-world engineering problems and building reliable, high-quality software solutions.

EDUCATION

BMS Institute of Technology and Management B.E. in Computer Science & Engineering- 9.03 CGPA	2023 – 2027
Sri Maruthi PU College Class XII (Computer Science) – 94.6%	2022 – 2023
Anugraha High School Class X – 79.3%	2020 – 2021

PROJECTS

Ocean Risk Vessel Monitoring System (ORVMS) 🌊

- **Tech Stack:** Python, Streamlit, Folium, Pandas, NumPy, SQLite, Scikit-learn, Shapely.
- Developed an integrated Maritime Situational Awareness platform that combines vessel tracking, geospatial visualization, OCR-based document processing, weather monitoring and intelligent alert generation into a unified dashboard.
- Implemented AIS data simulation and Isolation Forest-based anomaly detection using vessel attributes such as speed, latitude, longitude, heading, course and proximity to maritime boundaries to identify suspicious vessel behavior.
- Built backend modules for data processing, alert generation and SQLite-based data management, enabling efficient monitoring and analysis of maritime activities.
- Integrated multiple maritime intelligence features into a single platform, reducing the need to switch between separate systems for visualization, document analysis, weather monitoring and vessel surveillance.

Argo Base FloatChat AI

- **Tech Stack:** Python, PostgreSQL, Pandas, NumPy, Xarray, NetCDF4, Streamlit, REST APIs, Vector Database.
- Developed an AI-powered conversational platform for exploring large-scale oceanographic datasets by enabling natural language interaction with ARGO float data.
- Designed and implemented a data ingestion pipeline to automate the collection, parsing and conversion of complex NetCDF and Markdown datasets into structured database records for efficient querying.
- Built backend components including chatbot integration, prompt engineering, database management and visualization modules to simplify analysis of oceanographic data for non-technical users.
- Integrated APIs and data retrieval pipelines to generate context-aware responses and interactive visualizations from processed scientific datasets.

AURA-X Autonomous Unmanned Research Aircraft without GPS-Navigation System

- **Tech Stack:** ROS, Python, C++, OpenCV or B-SLAM3, Gazebo, PX4, MATLAB, TensorFlow.
 - Designing and developing an autonomous reconnaissance UAV capable of reliable GPS-denied navigation using Visual-Inertial Odometry (VIO) and sensor fusion techniques, which helps in the prevention of spoofing, jamming, EM warfare.
 - Developing intelligent path planning, obstacle avoidance and trajectory optimization using algorithms including A*, RRT, RRT*, PSO and APF for autonomous mission execution.
 - Researching real-time localization, mapping and autonomous flight control within a modular UAV architecture for robust navigation in challenging environments.
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TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript and SQL.

Web Technologies: HTML, CSS, JavaScript, React.js, REST APIs.

Databases: PostgreSQL, MongoDB, SQLite, Vector Databases.

Artificial Intelligence and Data Engineering: Machine Learning, Predictive Modeling, Anomaly Detection, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Data Processing, Data Visualization, Feature Engineering.

Autonomous Systems and Robotics: Autonomous Navigation, Path Planning, Sensor Fusion, Visual-Inertial Odometry (VIO).

Tools & Technologies: Git, GitHub, Docker, Linux, Streamlit, Folium, ROS, PX4, Gazebo, MATLAB, OpenCV, SolidWorks, Pandas, NumPy, Xarray, NetCDF4.

ACHIEVEMENTS

- Founder & Vice President, ASTRA Club, BMSIT&M.
- Secured 2nd Place in InCSEption Inter-College Hackathon.
- Received the Track Prize at ANVESHANA National-Level Hackathon.